# **The misdiagnosed mediator:** Estimating the effect of maternal age on preterm birth risk in the presence of misclassified gestational hypertension

**Kimberly A. H. Webb** and Martin T. Wells

International Day of Women in Statistics and Data Science

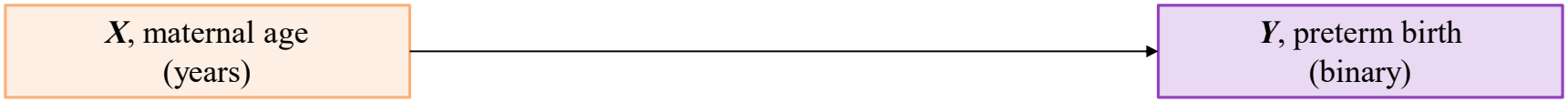
October 14, 2025

# Mediation analysis

- **Mediation analysis** quantifies the effect of an **exposure (X)** on an **outcome (Y)**, mediated by some **intermediate (M)**.

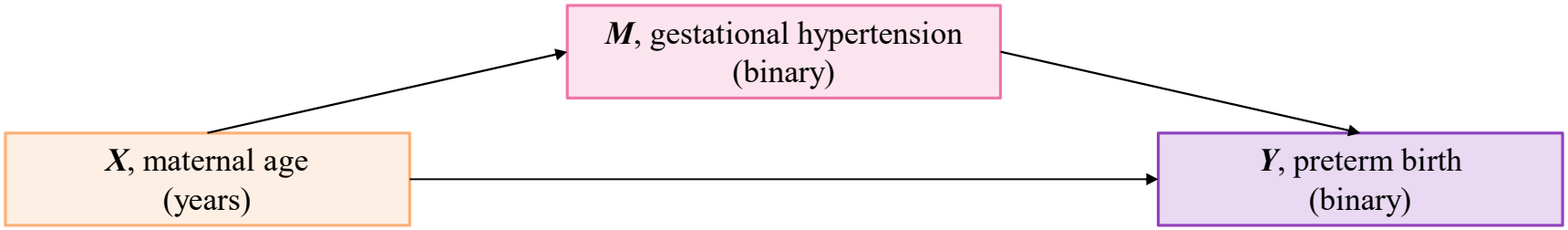
# Mediation analysis

- **Mediation analysis** quantifies the effect of an **exposure (X)** on an **outcome (Y)**, mediated by some **intermediate (M)**.



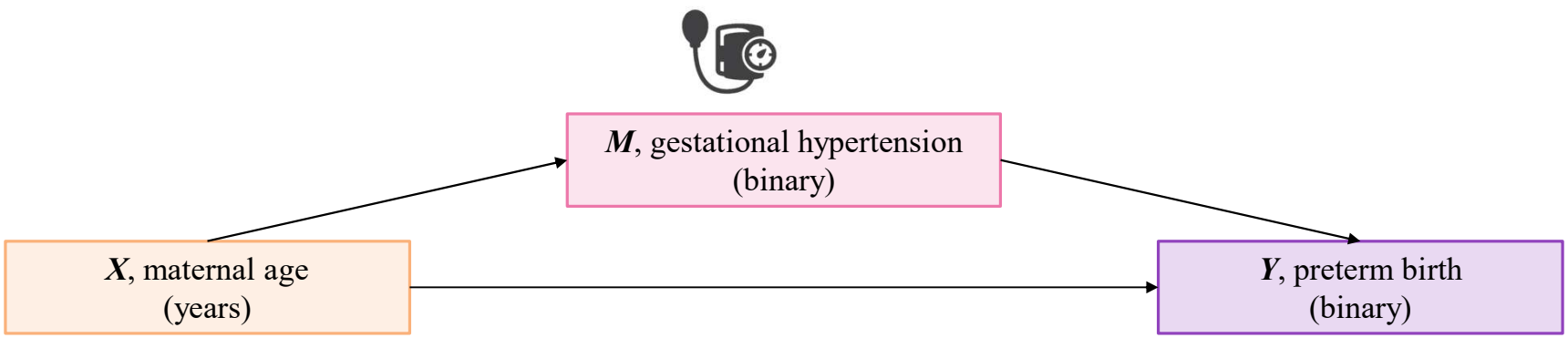
# Mediation analysis

- **Mediation analysis** quantifies the effect of an **exposure (X)** on an **outcome (Y)**, mediated by some **intermediate (M)**.



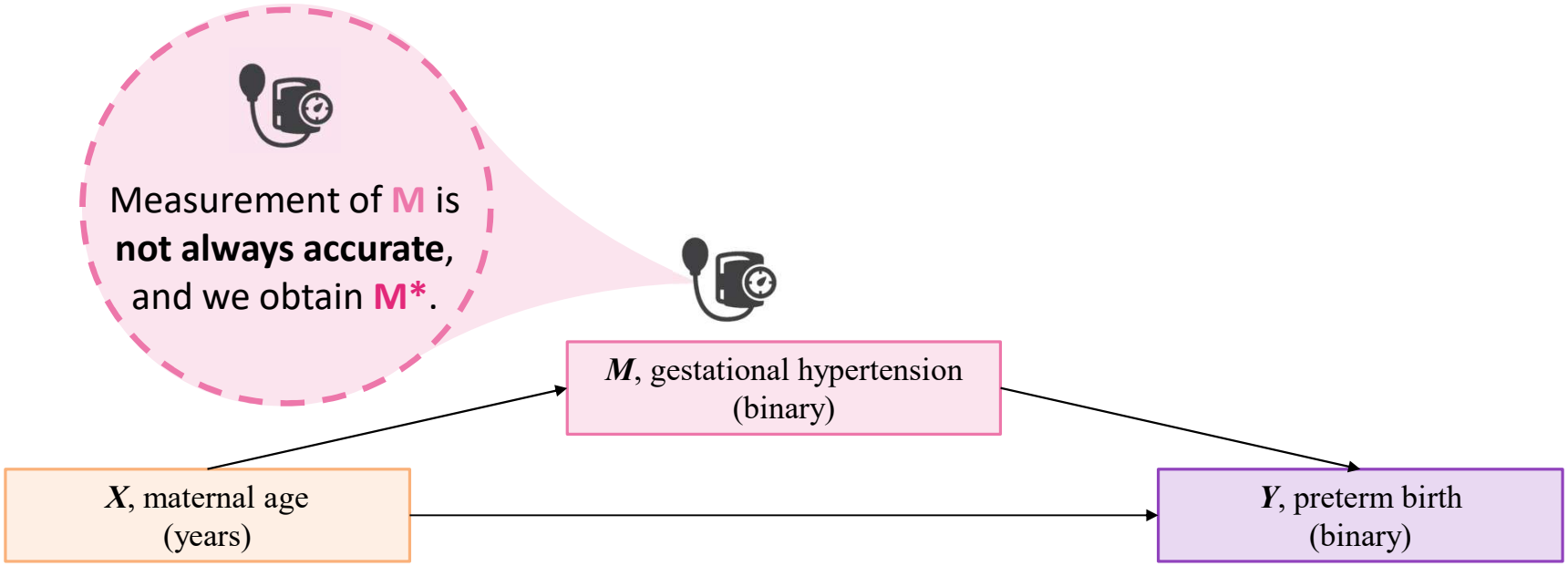
# Mediation analysis

- **Mediation analysis** quantifies the effect of an **exposure (X)** on an **outcome (Y)**, mediated by some **intermediate (M)**.



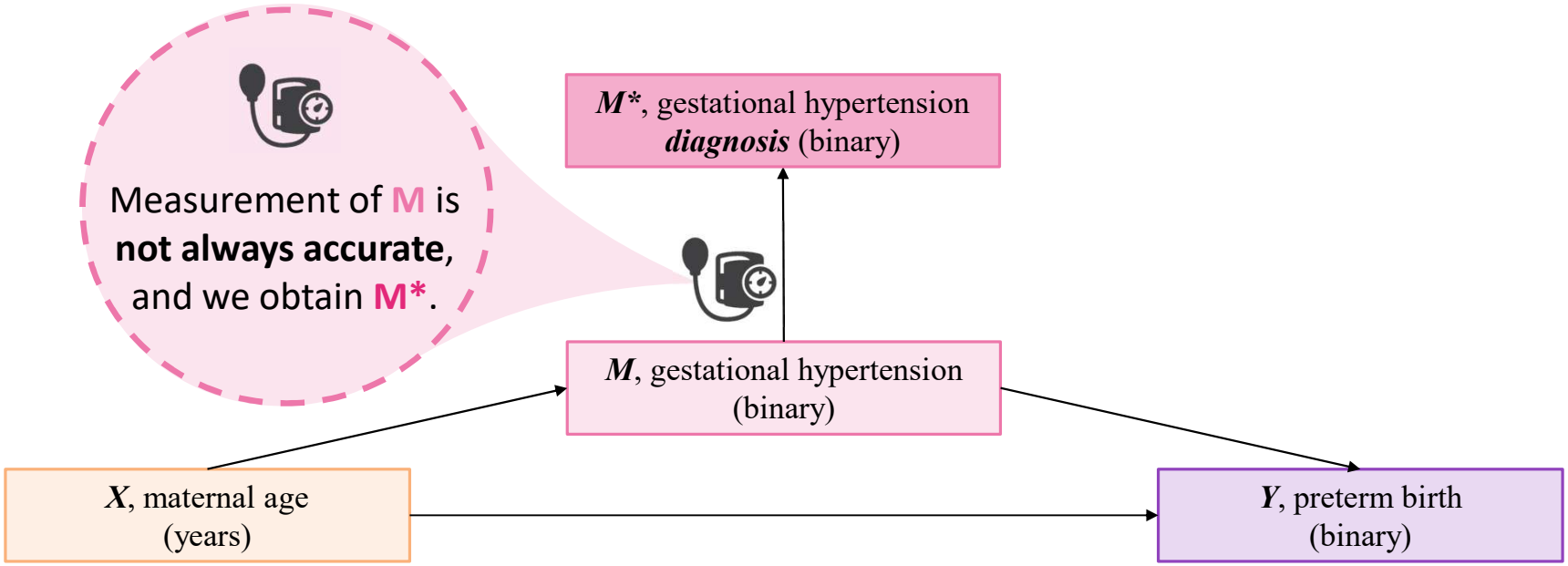
# Mediation analysis

- **Mediation analysis** quantifies the effect of an **exposure (X)** on an **outcome (Y)**, mediated by some **intermediate (M)**.



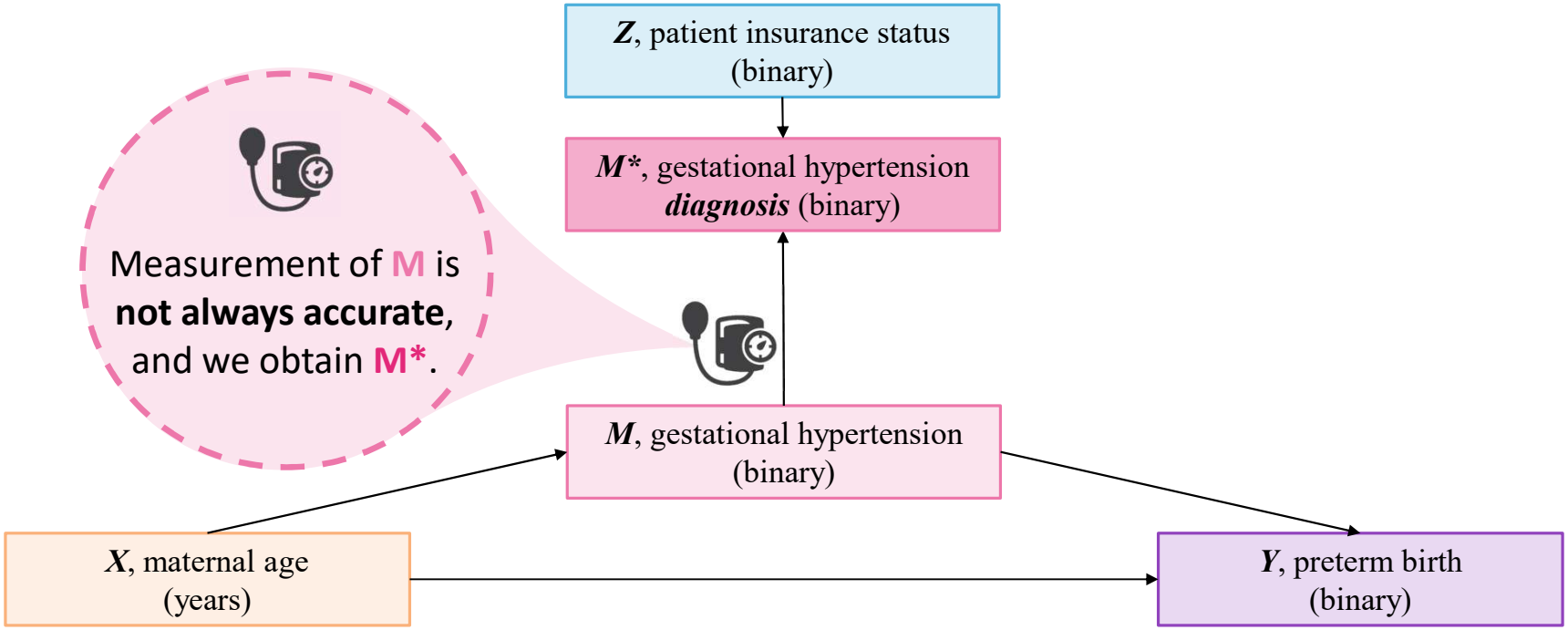
# Mediation analysis

- **Mediation analysis** quantifies the effect of an **exposure (X)** on an **outcome (Y)**, mediated by some **intermediate (M)**.



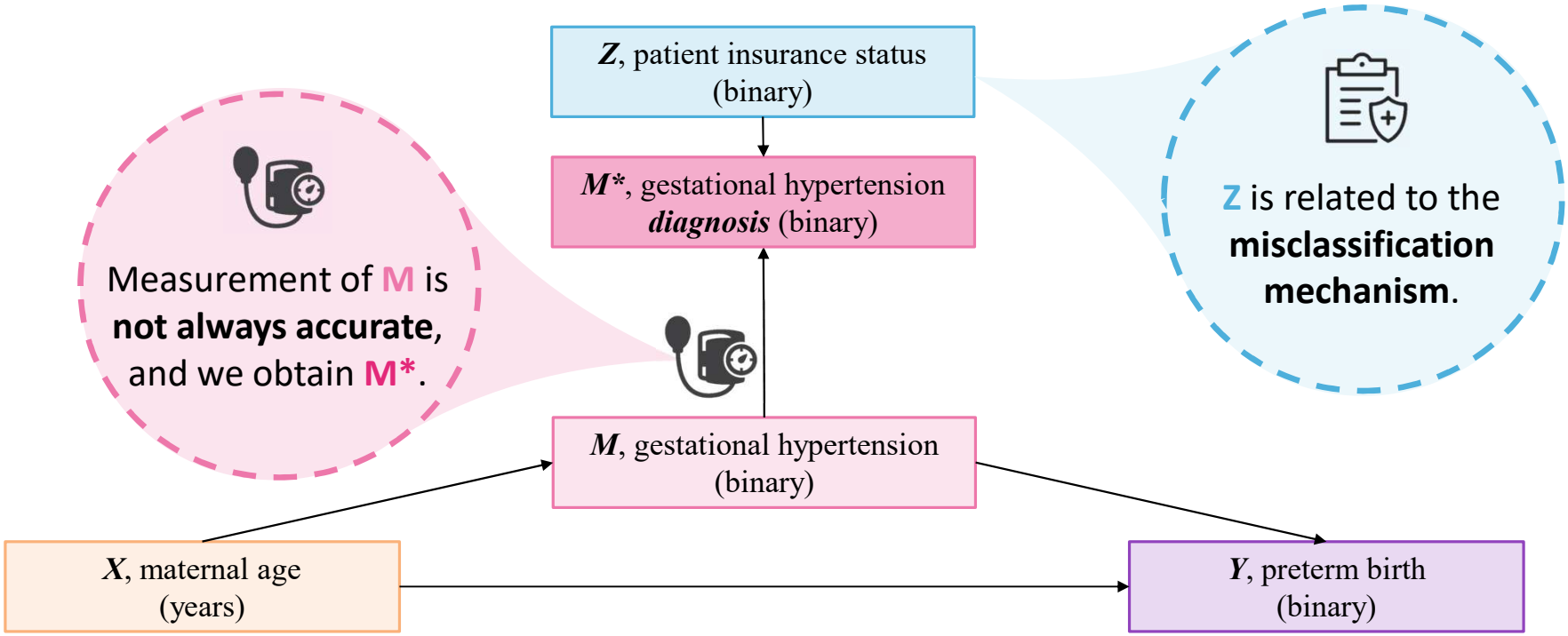
# Mediation analysis

- **Mediation analysis** quantifies the effect of an **exposure (X)** on an **outcome (Y)**, mediated by some **intermediate (M)**.

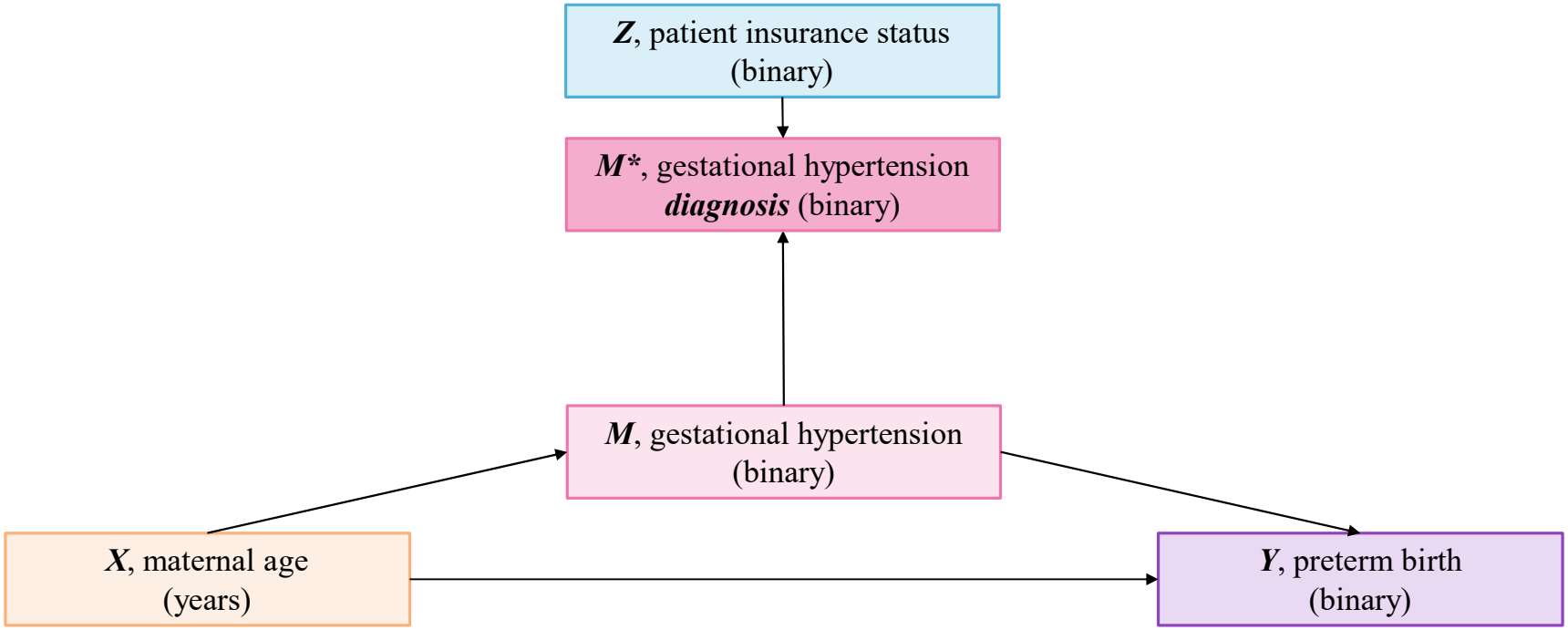


# Mediation analysis

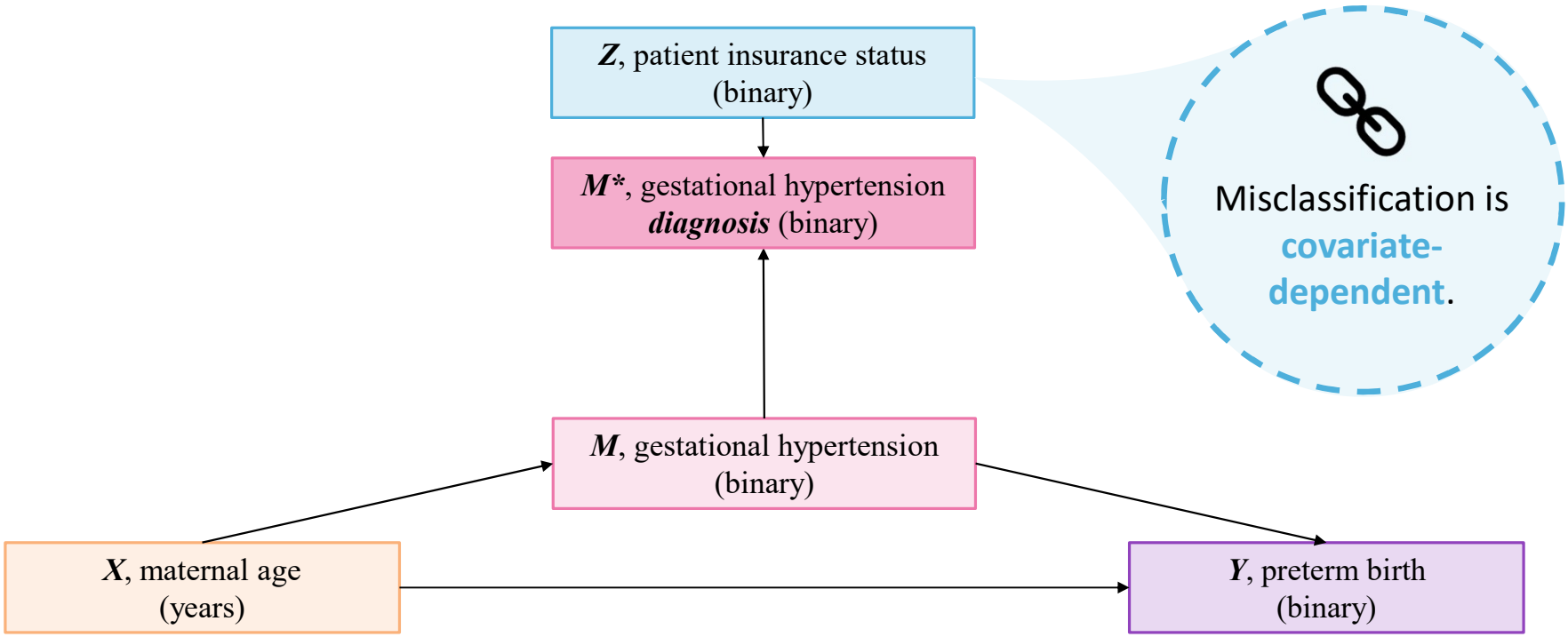
- **Mediation analysis** quantifies the effect of an **exposure (X)** on an **outcome (Y)**, mediated by some **intermediate (M)**.



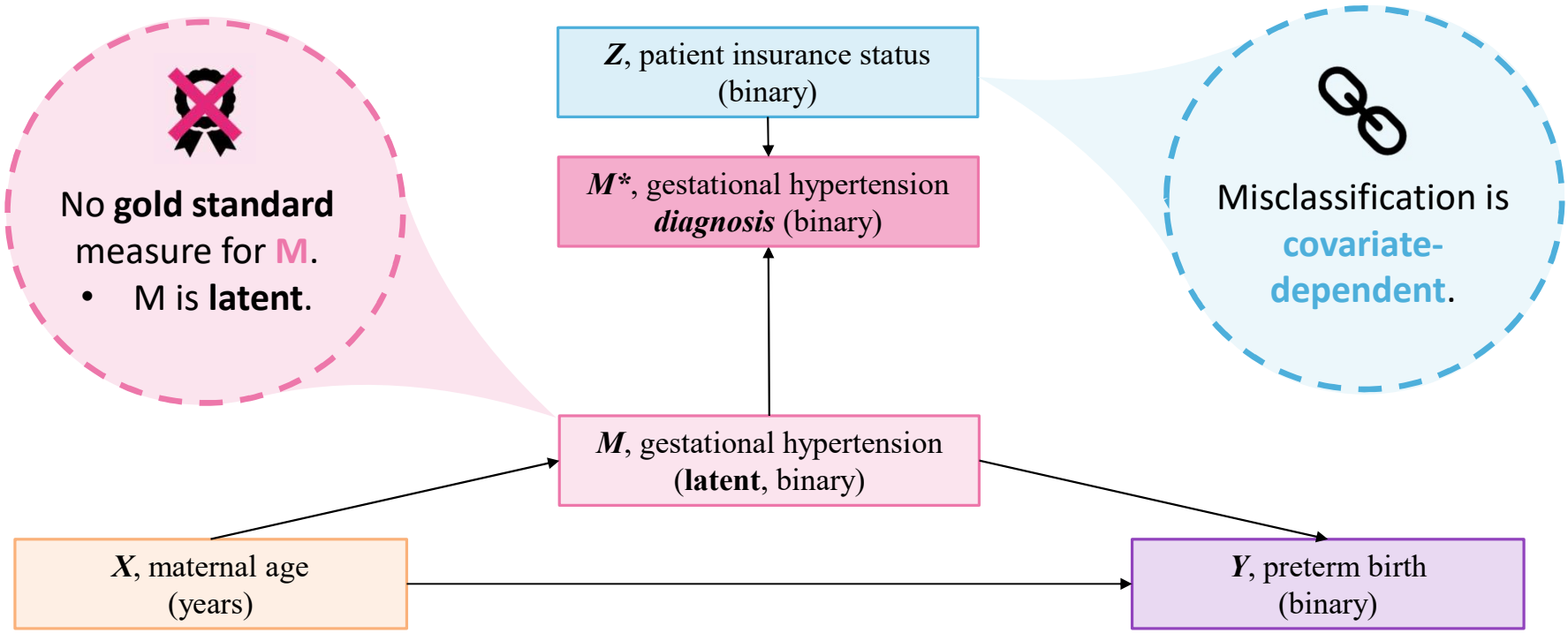
# Our analysis is complicated by misdiagnosis in *M*



# Our analysis is complicated by misdiagnosis in *M*



# Our analysis is complicated by misdiagnosis in *M*

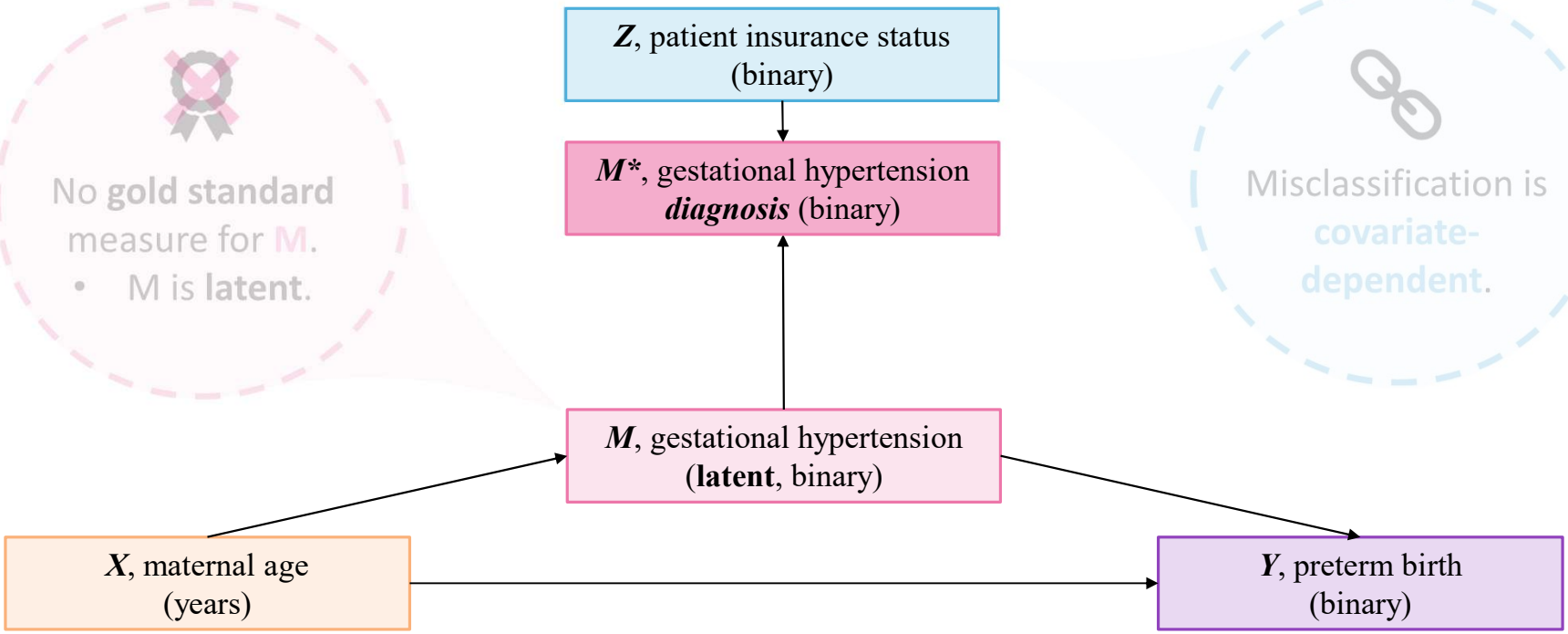


# Our analysis is complicated by misdiagnosis in *M*

Ignoring misclassification in M* → Bias in parameter and effect estimates

No **gold standard** measure for **M**.
- M is **latent**.

Misclassification is **covariate-dependent**.

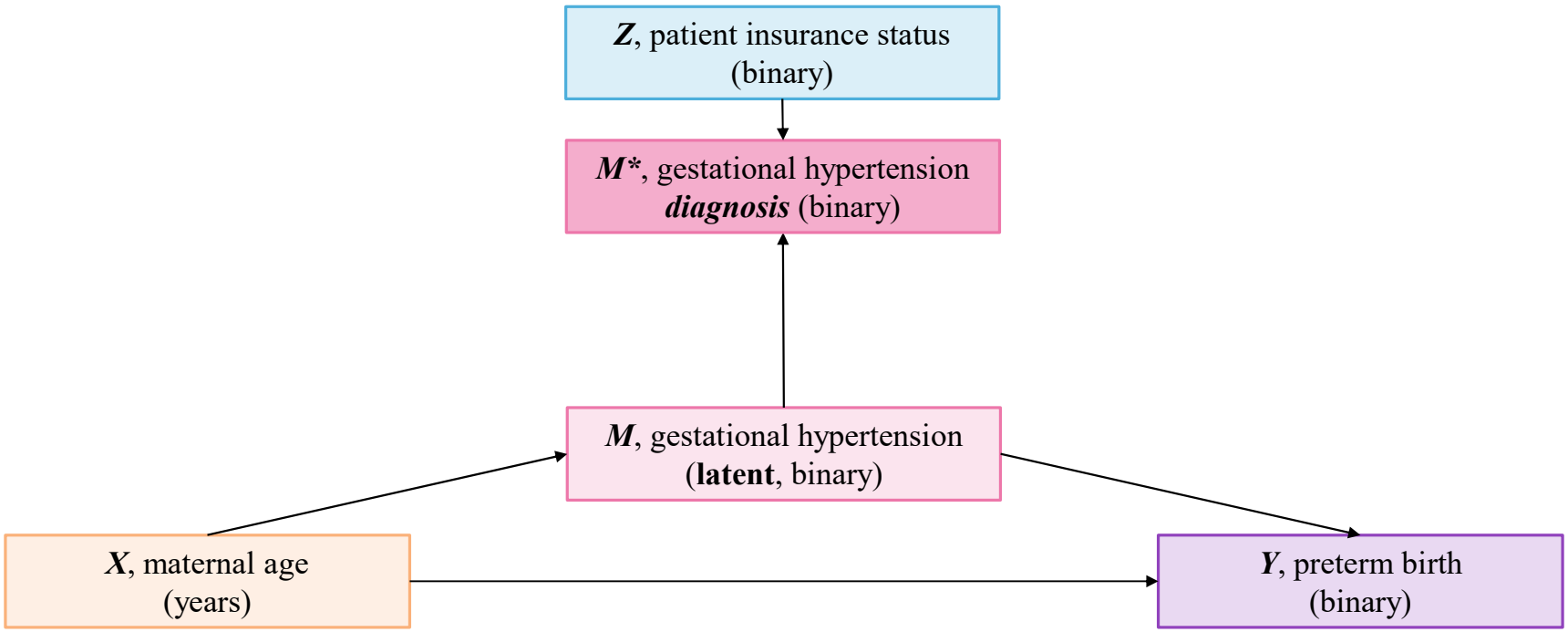


# Analysis Plan

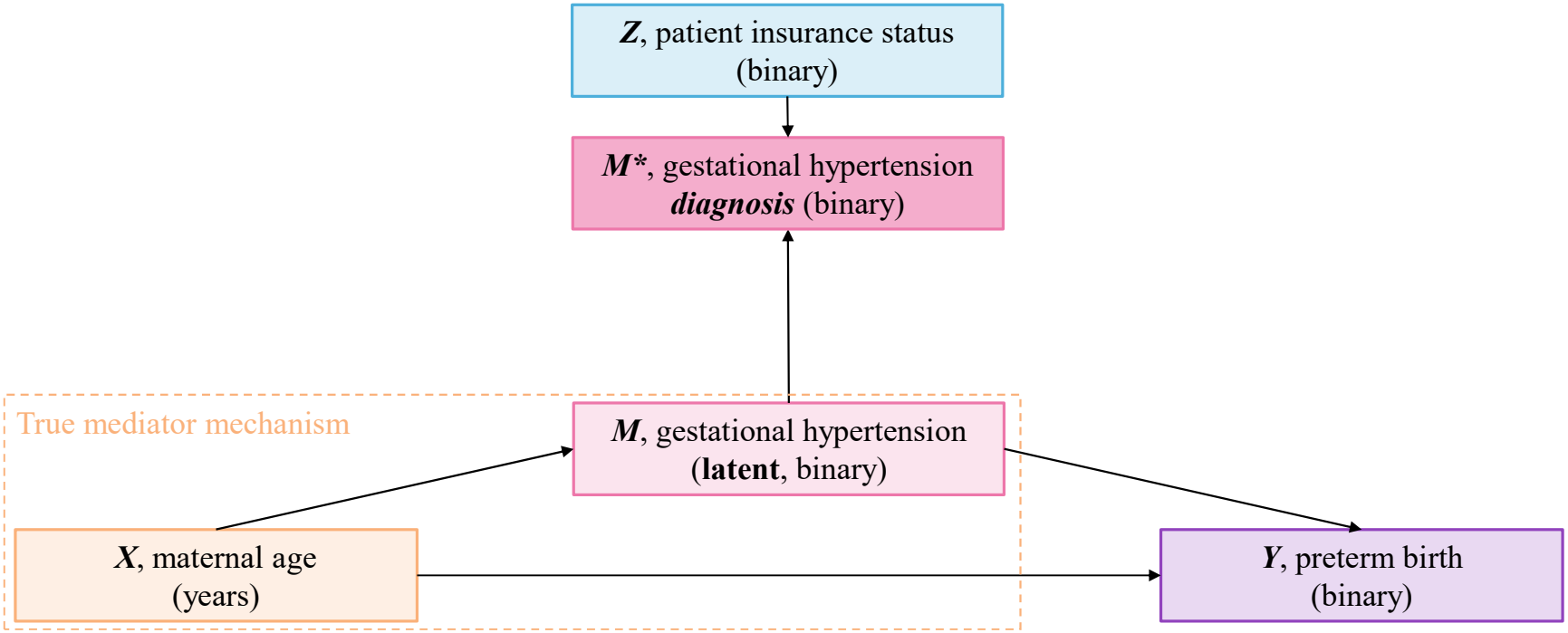
1. Define the **misclassification model**
2. Develop **estimation methods** for parameters of interest
3. Use parameter estimates to compute **(in)direct effects and misclassification rates**

1

# Our model accounts for misdiagnosis in *M*

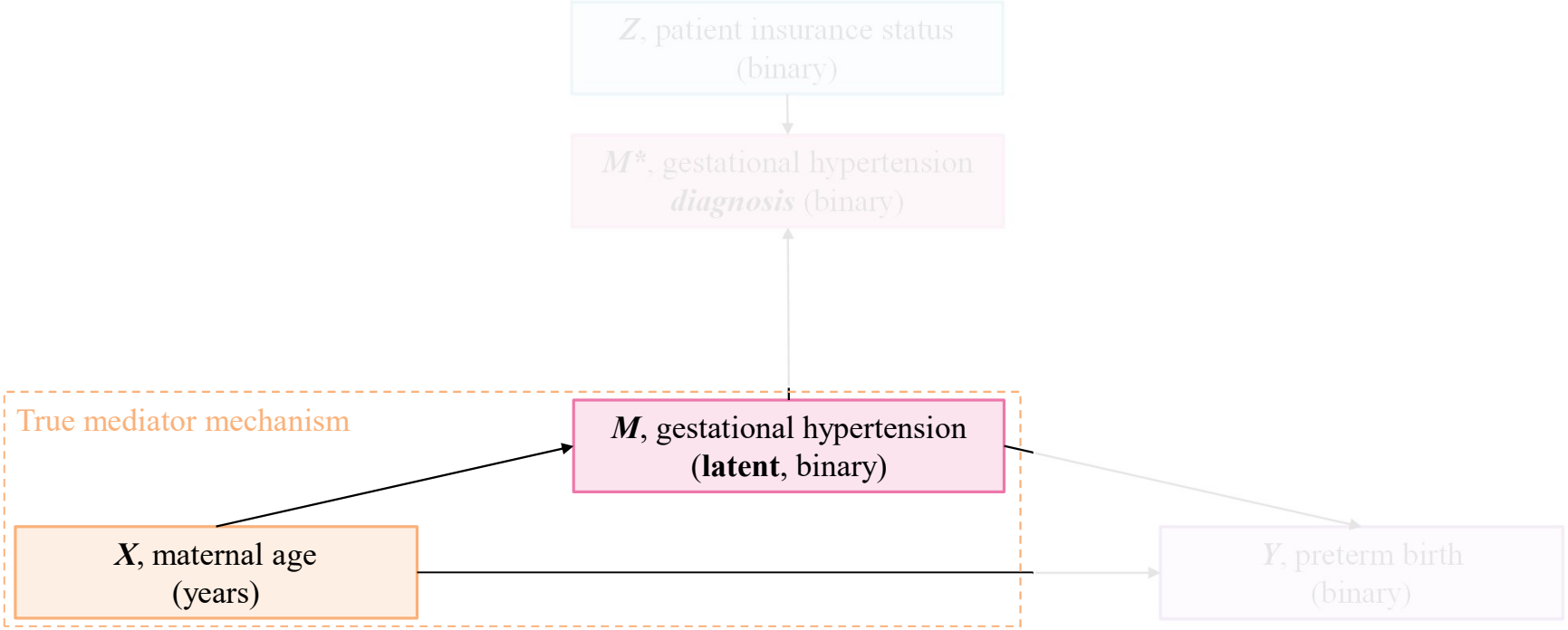


# Our model accounts for misdiagnosis in *M*



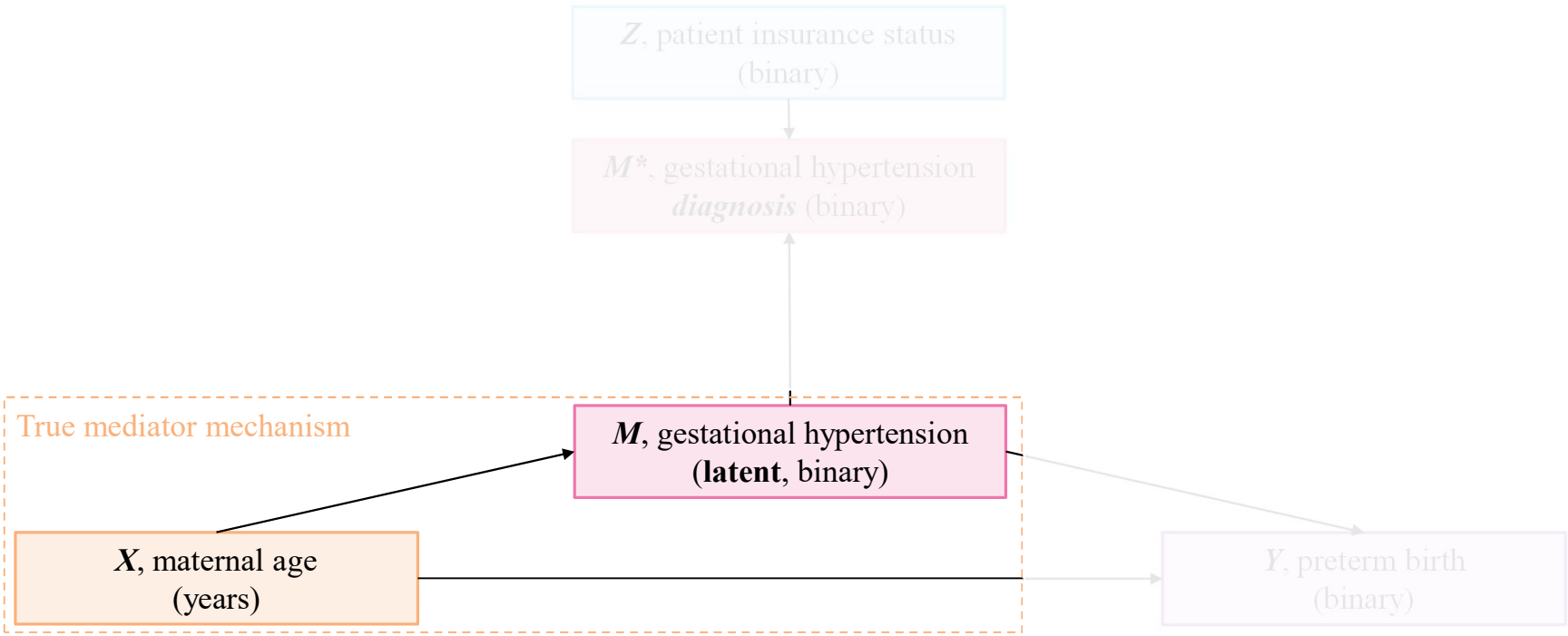
1

# Our model accounts for misdiagnosis in *M*



**True mediator mechanism:** $\text{logit}\{P(M = 1|X, \boldsymbol{C}; \boldsymbol{\beta})\} = \beta_0 + \beta_X X + \boldsymbol{\beta}_C \boldsymbol{C}$

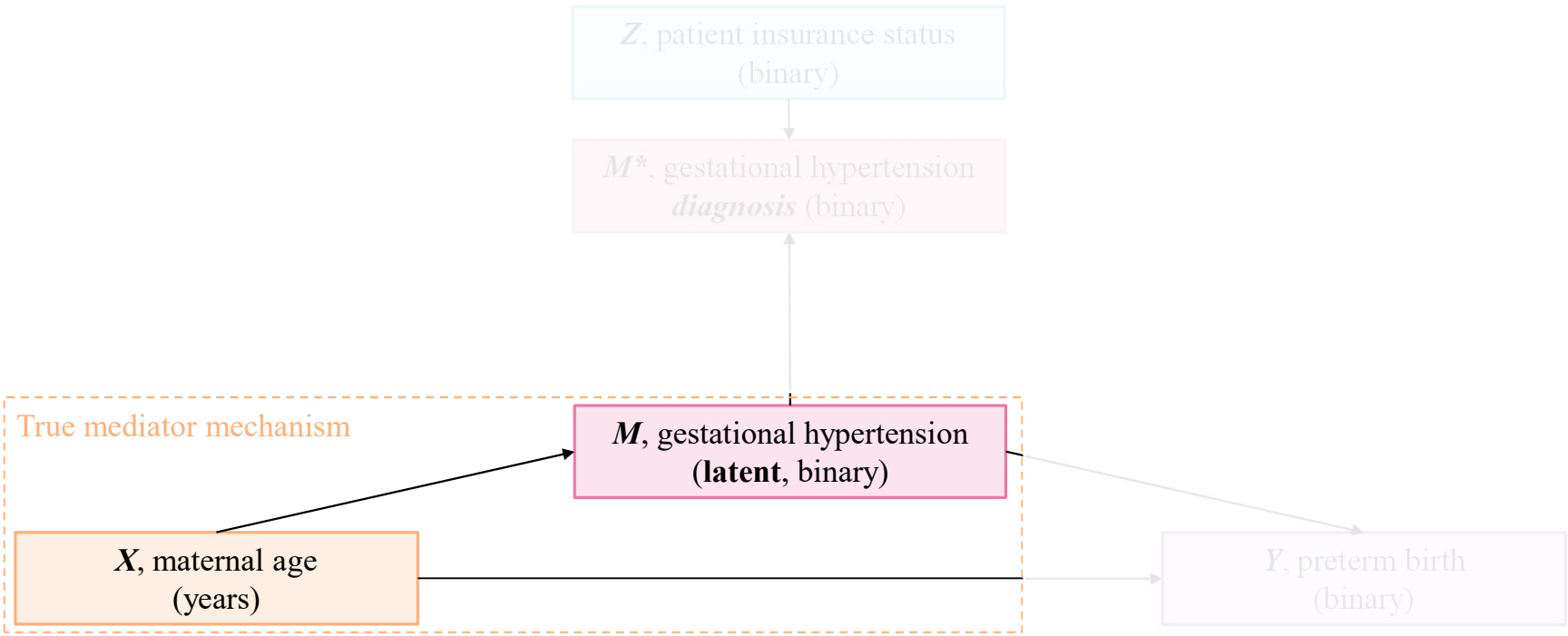
1



**True mediator mechanism:** $\text{logit}\{P(M = 1|X, \boldsymbol{C}; \boldsymbol{\beta})\} = \beta_0 + \beta_X X + \boldsymbol{\beta_C C}$

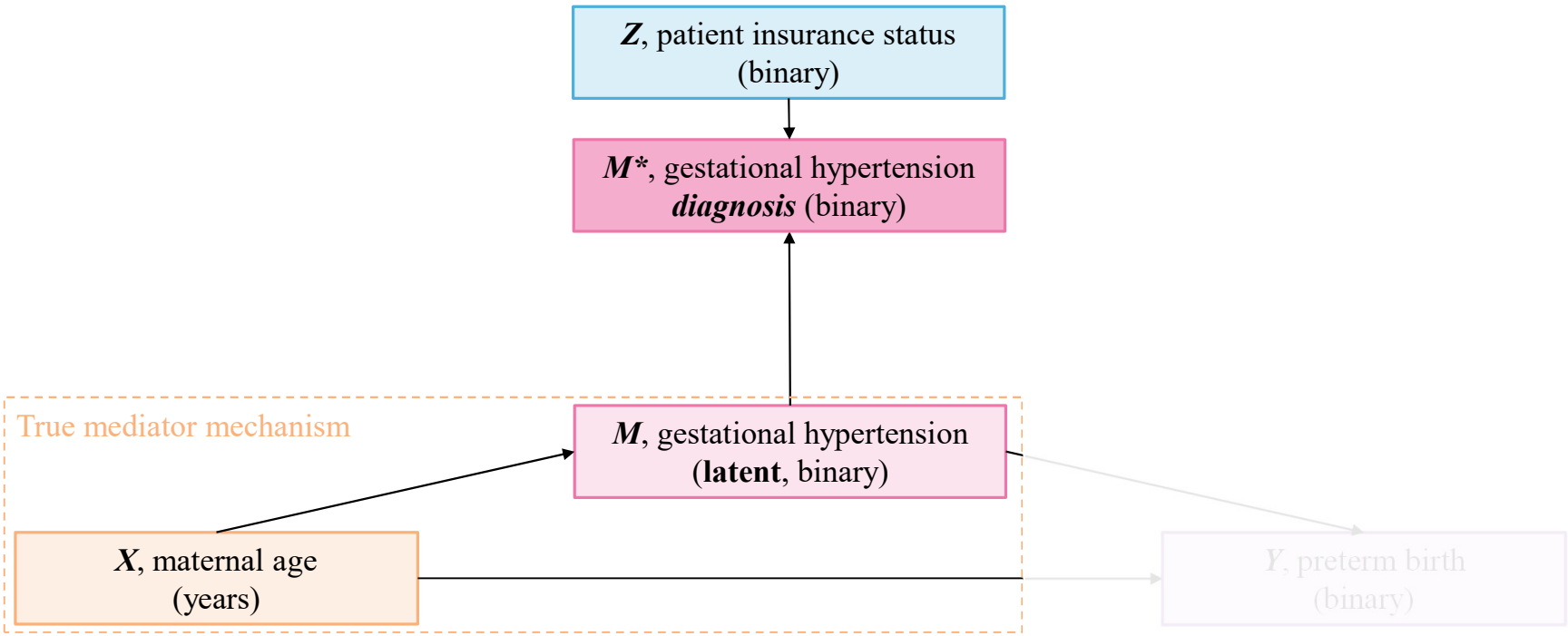
1

***C***, covariates



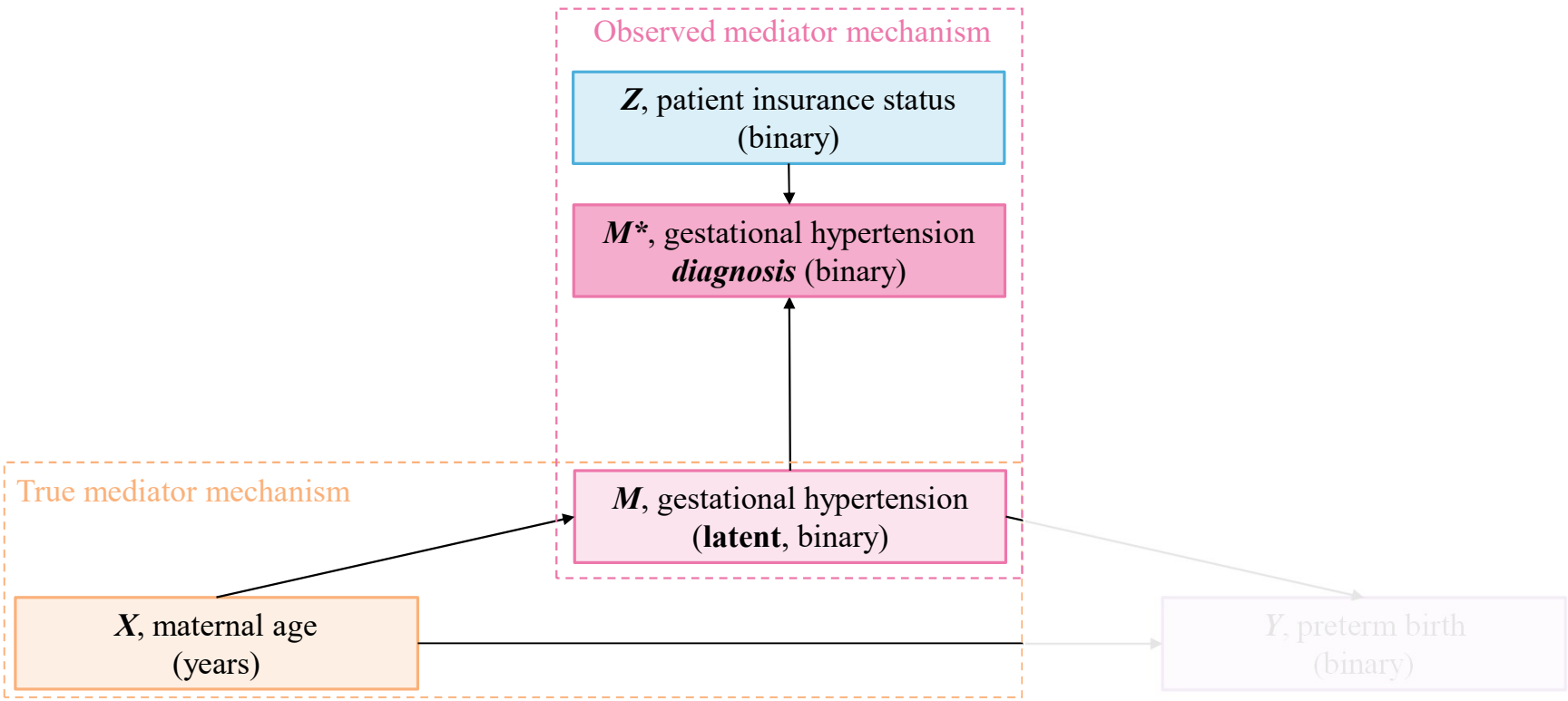
**True mediator mechanism:** $\text{logit}\{P(M = 1|X, \boldsymbol{C}; \boldsymbol{\beta})\} = \beta_0 + \beta_X X + \boldsymbol{\beta}_C \boldsymbol{C}$

1



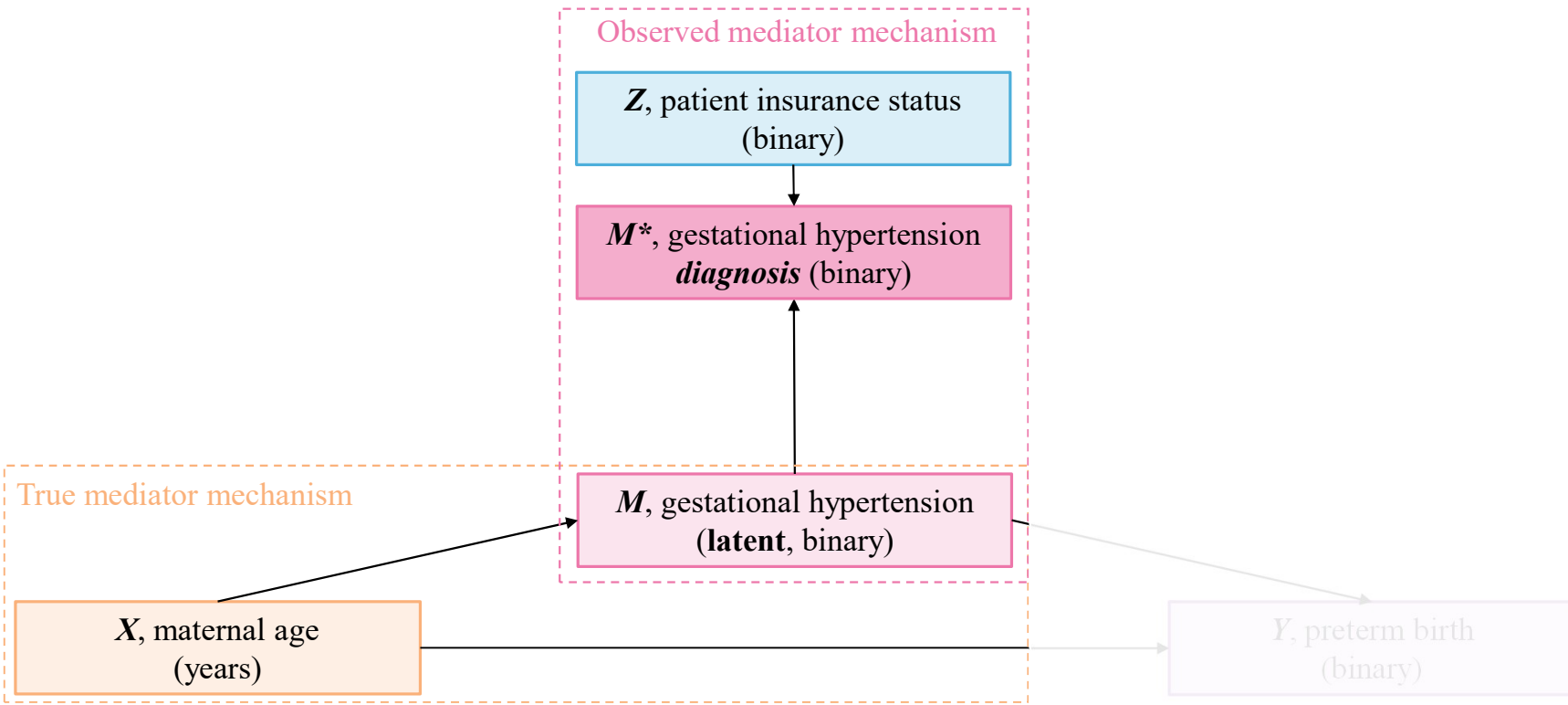
**True mediator mechanism:** $\text{logit}\{P(M = 1|X, \boldsymbol{C}; \boldsymbol{\beta})\} = \beta_0 + \beta_X X + \boldsymbol{\beta}_C \boldsymbol{C}$

1



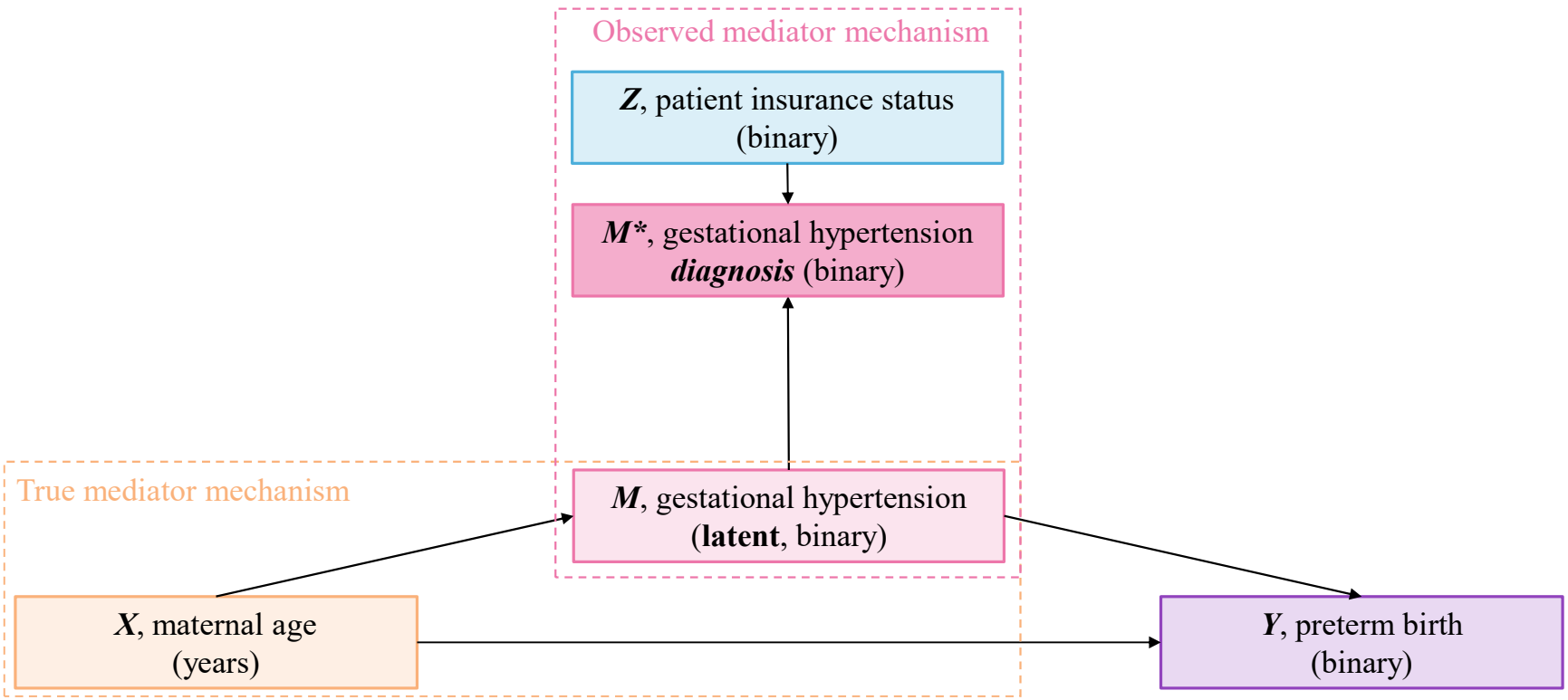
**True mediator mechanism:** $\text{logit}\{P(M = 1|X, \boldsymbol{C}; \boldsymbol{\beta})\} = \beta_0 + \beta_X X + \boldsymbol{\beta_C C}$ 1

**Observed mediator mechanism:** $\text{logit}\{P(M^* = 1|M = m, \boldsymbol{Z}; \boldsymbol{\gamma})\} = \gamma_{1m0} + \boldsymbol{\gamma_{1mZ} Z}$



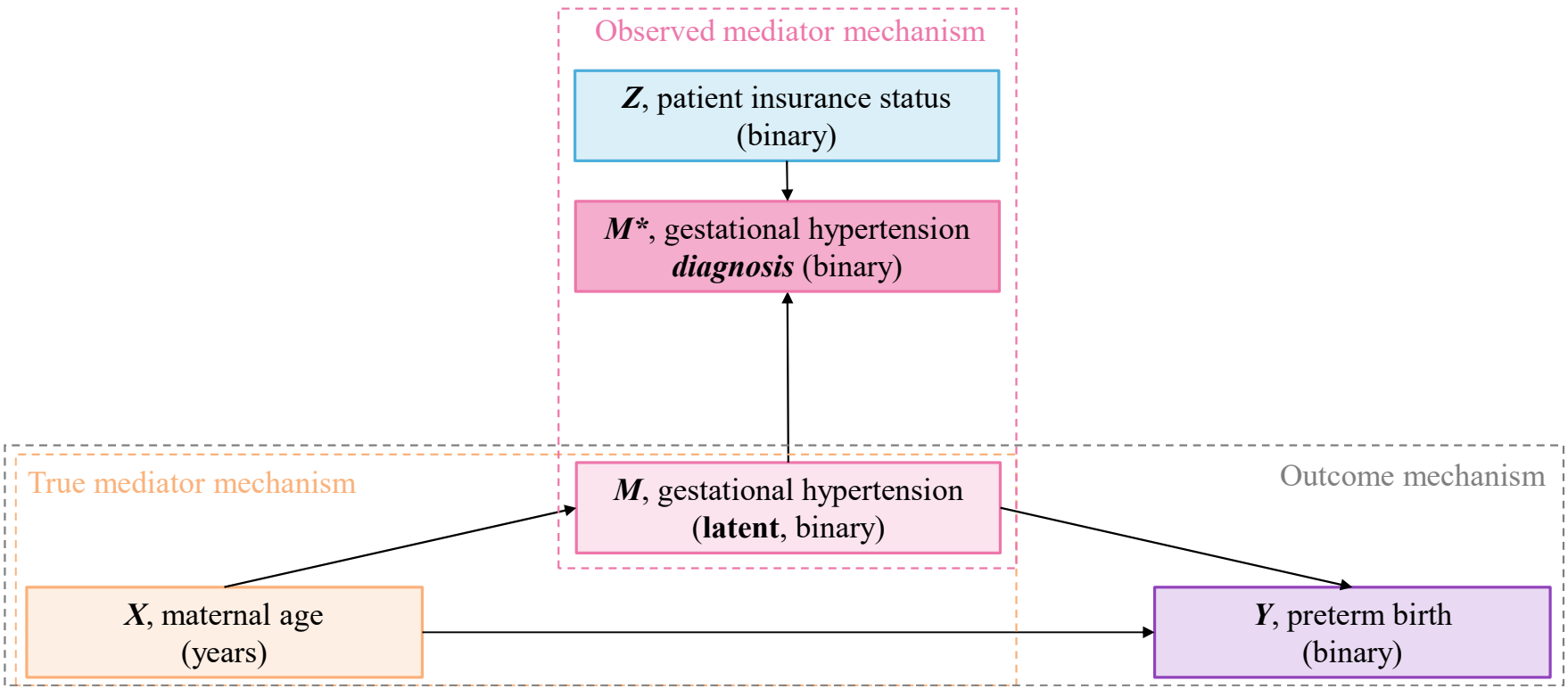
**True mediator mechanism:** $\text{logit}\{P(M = 1|X, \boldsymbol{C}; \boldsymbol{\beta})\} = \beta_0 + \beta_X X + \boldsymbol{\beta_C C}$ (1)

**Observed mediator mechanism:** $\text{logit}\{P(M^* = 1|M = m, \boldsymbol{Z}; \boldsymbol{\gamma})\} = \gamma_{1m0} + \boldsymbol{\gamma_{1mZ} Z}$



**True mediator mechanism:** $\text{logit}\{P(M = 1|X, \boldsymbol{C}; \boldsymbol{\beta})\} = \beta_0 + \beta_X X + \boldsymbol{\beta_C C}$ 1

**Observed mediator mechanism:** $\text{logit}\{P(M^* = 1|M = m, \boldsymbol{Z}; \boldsymbol{\gamma})\} = \gamma_{1m0} + \boldsymbol{\gamma_{1mZ}} \boldsymbol{Z}$

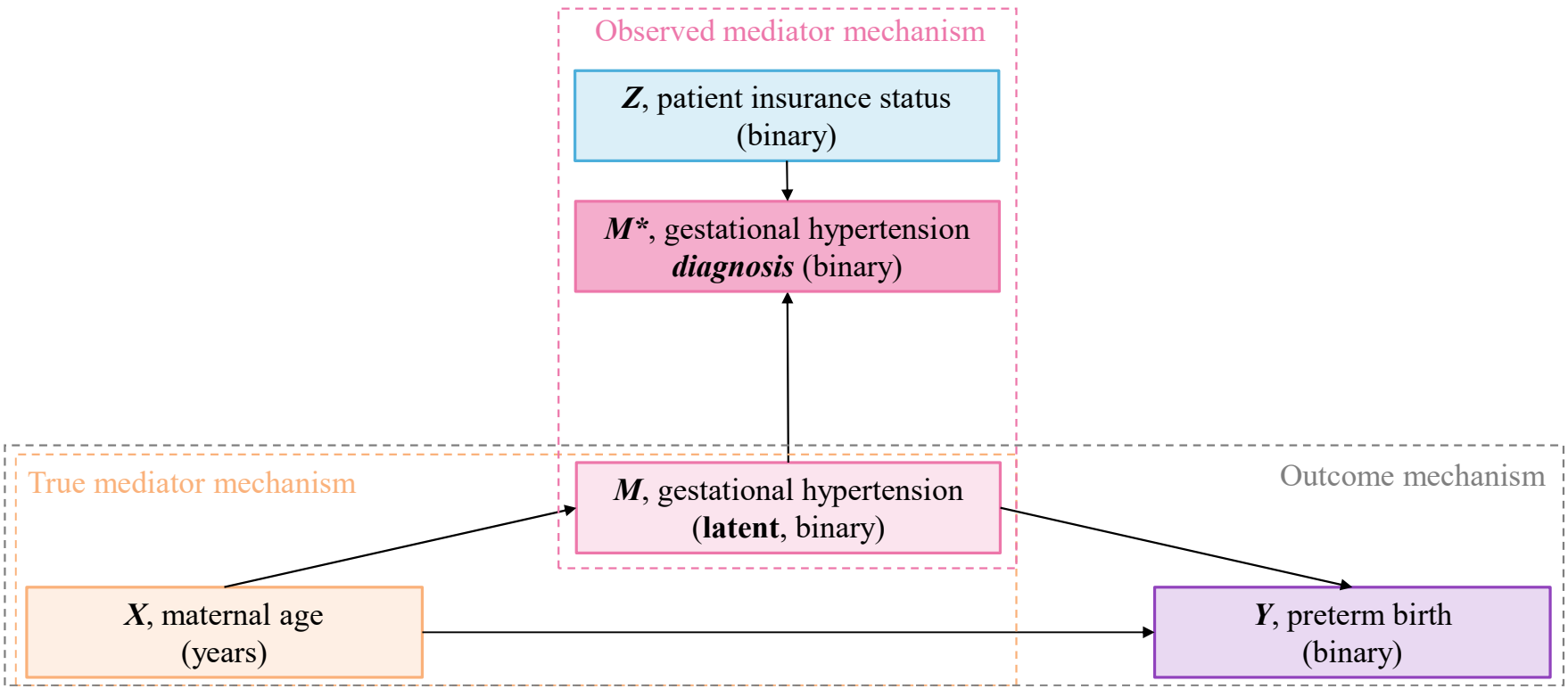


1

**True mediator mechanism:** $\text{logit}\{P(M = 1|X, \boldsymbol{C}; \boldsymbol{\beta})\} = \beta_0 + \beta_X X + \boldsymbol{\beta_C C}$

**Observed mediator mechanism:** $\text{logit}\{P(M^* = 1|M = m, \boldsymbol{Z}; \boldsymbol{\gamma})\} = \gamma_{1m0} + \boldsymbol{\gamma_{1mZ} Z}$

**Outcome mechanism:** $E(Y|X, \boldsymbol{C}, M) = \theta_0 + \theta_X X + \boldsymbol{\theta_C C} + \theta_M M + \theta_{XM} XM$



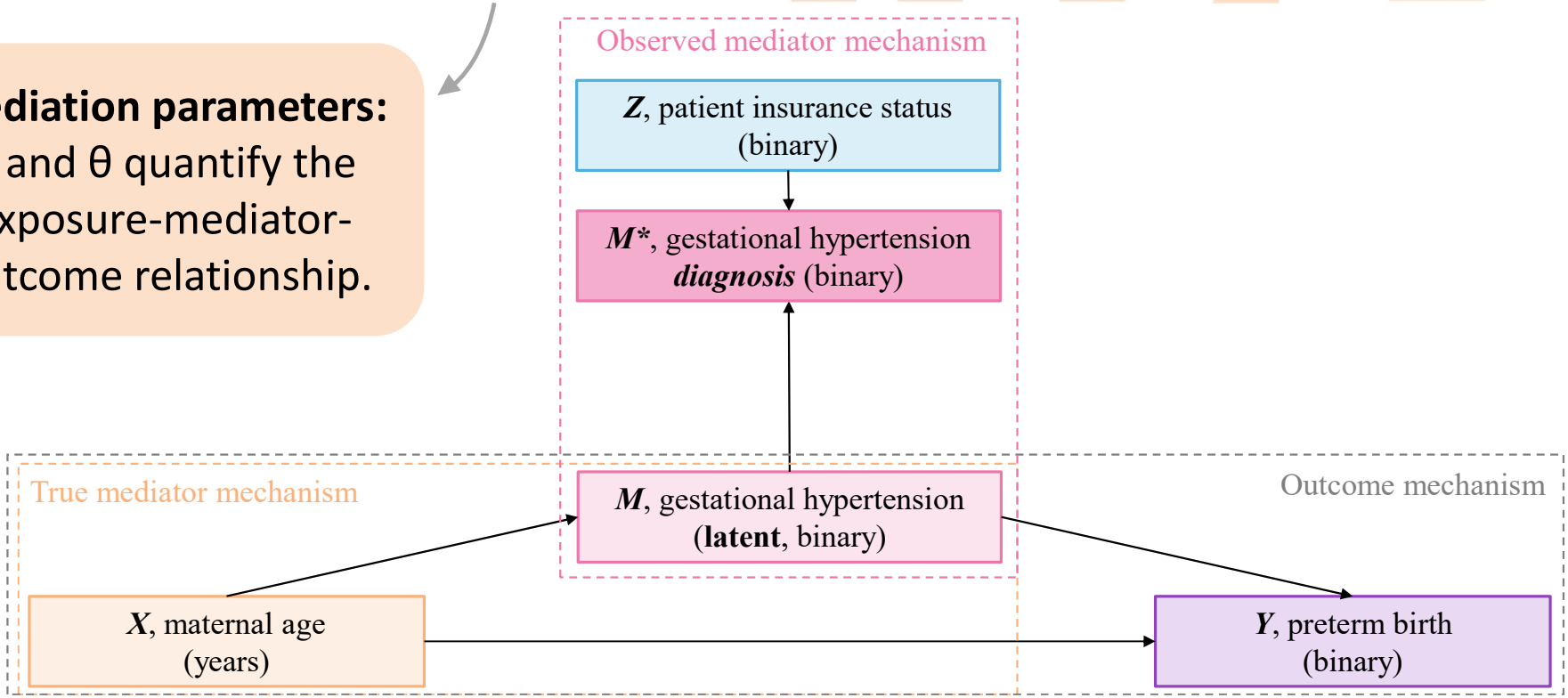
1

**True mediator mechanism:** $\text{logit}\{P(M = 1|X, \boldsymbol{C}; \boldsymbol{\beta})\} = \beta_0 + \beta_X X + \boldsymbol{\beta_C C}$

**Observed mediator mechanism:** $\text{logit}\{P(M^* = 1|M = m, \boldsymbol{Z}; \boldsymbol{\gamma})\} = \gamma_{1m0} + \boldsymbol{\gamma_{1mZ} Z}$

**Outcome mechanism:** $E(Y|X, \boldsymbol{C}, M) = \theta_0 + \theta_X X + \boldsymbol{\theta_C C} + \theta_M M + \theta_{XM} XM$

**Mediation parameters:**
β and θ quantify the exposure-mediator-outcome relationship.



1

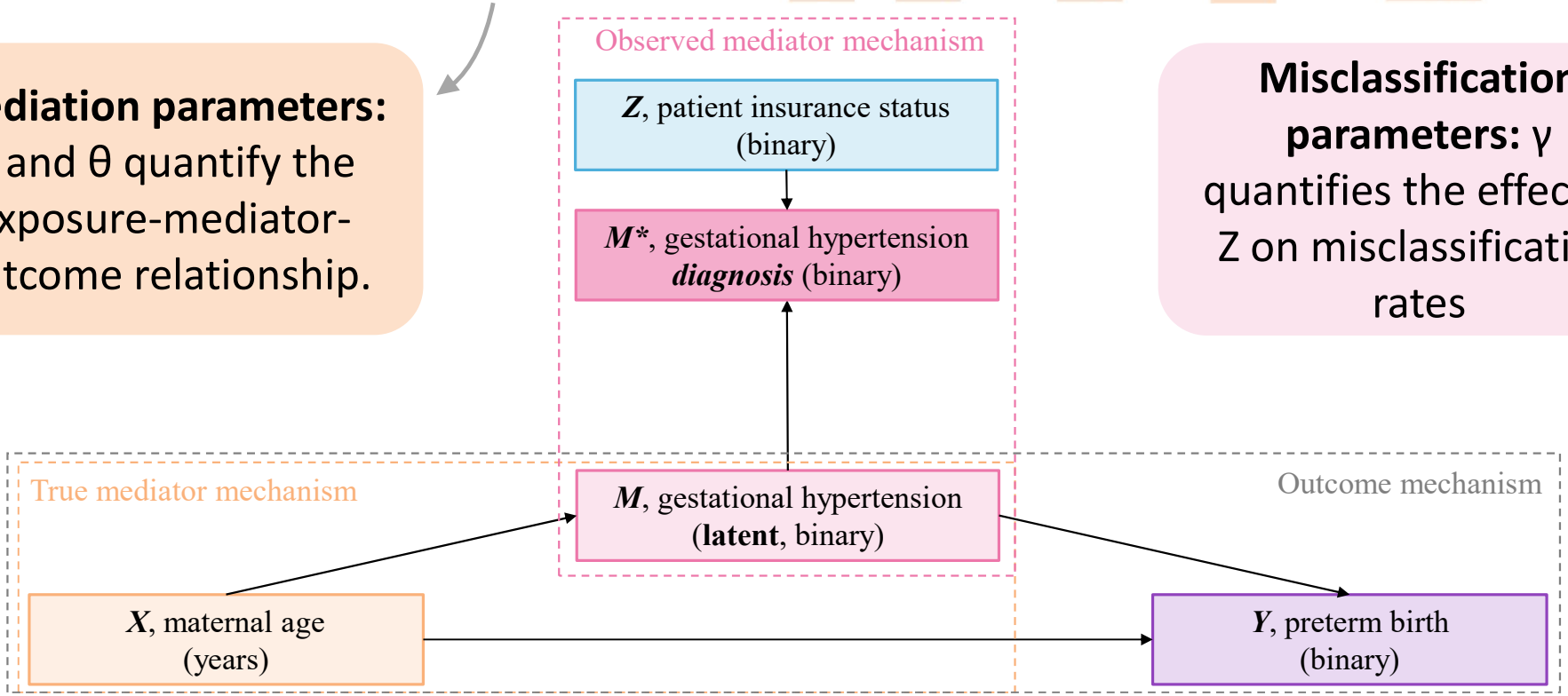
**True mediator mechanism:** $\text{logit}\{P(M=1|X,\boldsymbol{C};\boldsymbol{\beta})\} = \beta_0 + \beta_X X + \boldsymbol{\beta_C}\boldsymbol{C}$

**Observed mediator mechanism:** $\text{logit}\{P(M^*=1|M=m,\boldsymbol{Z};\boldsymbol{\gamma})\} = \gamma_{1m0} + \boldsymbol{\gamma_{1mZ}}\boldsymbol{Z}$

**Outcome mechanism:** $E(Y|X,\boldsymbol{C},M) = \theta_0 + \theta_X X + \boldsymbol{\theta_C}\boldsymbol{C} + \theta_M M + \theta_{XM} XM$

**Mediation parameters:** β and θ quantify the exposure-mediator-outcome relationship.

**Misclassification parameters:** γ quantifies the effect of Z on misclassification rates



2

# We developed 3 estimation methods

**True mediator mechanism:** $\text{logit}\{P(M=1|X,\boldsymbol{C};\boldsymbol{\beta})\} = \beta_0 + \beta_X X + \boldsymbol{\beta_C C}$

**Observed mediator mechanism:** $\text{logit}\{P(M^*=1|M=m,\boldsymbol{Z};\boldsymbol{\gamma})\} = \gamma_{1m0} + \boldsymbol{\gamma_{1mZ} Z}$

**Outcome mechanism:** $E(Y|X,\boldsymbol{C},M) = \theta_0 + \theta_X X + \boldsymbol{\theta_C C} + \theta_M M + \theta_{XM} XM$

| #1: OLS Correction | #2: Predictive value weighting | #3: An EM algorithm |
|---|---|---|

2

# We developed 3 estimation methods

**True mediator mechanism:** $\text{logit}\{P(M = 1|X, \boldsymbol{C}; \boldsymbol{\beta})\} = \beta_0 + \beta_X X + \boldsymbol{\beta_C C}$

**Observed mediator mechanism:** $\text{logit}\{P(M^* = 1|M = m, \boldsymbol{Z}; \boldsymbol{\gamma})\} = \gamma_{1m0} + \boldsymbol{\gamma_{1mZ} Z}$

**Outcome mechanism:** $E(Y|X, \boldsymbol{C}, M) = \theta_0 + \theta_X X + \boldsymbol{\theta_C C} + \theta_M M + \theta_{XM} XM$

| #1: OLS Correction | #2: Predictive value weighting | #3: An EM algorithm |
|---|---|---|

**Key point:** We can use ***COMBO*** to estimate subject-level sensitivity and specificity, and then plug these values into existing misclassification correction procedures.

- Existing procedures relied on *known* sensitivity and specificity.

1. Extended from Nguimkeu, Rosenman, and Tennekoon (2021), "Regression with a misclassified binary regressor: Correcting for hidden bias".
2. Extended from Lyles and Lin (2010), "Sensitivity analysis for misclassification in logistic regression via likelihood methods and PVW".

# We developed 3 estimation methods

**True mediator mechanism:** $\text{logit}\{P(M=1|X,\boldsymbol{C};\boldsymbol{\beta})\} = \beta_0 + \beta_X X + \boldsymbol{\beta_C C}$

**Observed mediator mechanism:** $\text{logit}\{P(M^*=1|M=m,\boldsymbol{Z};\boldsymbol{\gamma})\} = \gamma_{1m0} + \boldsymbol{\gamma_{1mZ}Z}$

**Outcome mechanism:** $E(Y|X,\boldsymbol{C},M) = \theta_0 + \theta_X X + \boldsymbol{\theta_C C} + \theta_M M + \theta_{XM} XM$

#1: OLS Correction | #2: Predictive value weighting | #3: An EM algorithm

**Complete data log-likelihood:**

$$\ell_{complete}(\boldsymbol{\beta},\boldsymbol{\gamma},\boldsymbol{\theta};X,\boldsymbol{C},\boldsymbol{Z},Y)$$
$$= \sum_{i=1}^{N}\left[\ell_{Y|X,M,\boldsymbol{C}}(\boldsymbol{\theta};X_i,M_i,\boldsymbol{C}_i,Y_i) + \sum_{j=1}^{2} m_{ij}\log\{\pi_{ij}\} + \sum_{j=1}^{2}\sum_{\ell=1}^{2} m_{ij}m^*_{i\ell}\log\{\pi^*_{i\ell j}\}\right]$$

# We developed 3 estimation methods

**True mediator mechanism:** $\text{logit}\{P(M=1|X,\boldsymbol{C};\boldsymbol{\beta})\} = \beta_0 + \beta_X X + \boldsymbol{\beta_C C}$

**Observed mediator mechanism:** $\text{logit}\{P(M^*=1|M=m,\boldsymbol{Z};\boldsymbol{\gamma})\} = \gamma_{1m0} + \boldsymbol{\gamma_{1mZ} Z}$

**Outcome mechanism:** $E(Y|X,\boldsymbol{C},M) = \theta_0 + \theta_X X + \boldsymbol{\theta_C C} + \theta_M M + \theta_{XM} XM$

#1: OLS Correction | #2: Predictive value weighting | #3: An EM algorithm

**Complete data log-likelihood:**

$$\ell_{complete}(\boldsymbol{\beta},\boldsymbol{\gamma},\boldsymbol{\theta};X,\boldsymbol{C},\boldsymbol{Z},Y)$$

$$= \sum_{i=1}^{N}\left[\ell_{Y|X,M,\boldsymbol{C}}(\boldsymbol{\theta};X_i,M_i,\boldsymbol{C}_i,Y_i) + \sum_{j=1}^{2} m_{ij}\log\{\pi_{ij}\} + \sum_{j=1}^{2}\sum_{\ell=1}^{2} m_{ij}m^*_{i\ell}\log\{\pi^*_{i\ell j}\}\right]$$

**Outcome**

# We developed 3 estimation methods

**True mediator mechanism:** $\text{logit}\{P(M=1|X,\boldsymbol{C};\boldsymbol{\beta})\} = \beta_0 + \beta_X X + \boldsymbol{\beta_C C}$

**Observed mediator mechanism:** $\text{logit}\{P(M^*=1|M=m,\boldsymbol{Z};\boldsymbol{\gamma})\} = \gamma_{1m0} + \boldsymbol{\gamma_{1mZ} Z}$

**Outcome mechanism:** $E(Y|X,\boldsymbol{C},M) = \theta_0 + \theta_X X + \boldsymbol{\theta_C C} + \theta_M M + \theta_{XM} XM$

#1: OLS Correction | #2: Predictive value weighting | #3: An EM algorithm

**Complete data log-likelihood:**

$$\ell_{complete}(\boldsymbol{\beta},\boldsymbol{\gamma},\boldsymbol{\theta};X,\boldsymbol{C},\boldsymbol{Z},Y)$$

$$= \sum_{i=1}^{N}\left[\ell_{Y|X,M,\boldsymbol{C}}(\boldsymbol{\theta};X_i,M_i,\boldsymbol{C}_i,Y_i) + \sum_{j=1}^{2} m_{ij}\log\{\pi_{ij}\} + \sum_{j=1}^{2}\sum_{\ell=1}^{2} m_{ij}m^*_{i\ell}\log\{\pi^*_{i\ell j}\}\right]$$

$\mathbb{I}(M_i = j)$ $P(M_i = j)$

Outcome True mediator

# We developed 3 estimation methods

**True mediator mechanism:** $\text{logit}\{P(M=1|X,\boldsymbol{C};\boldsymbol{\beta})\} = \beta_0 + \beta_X X + \boldsymbol{\beta_C C}$

**Observed mediator mechanism:** $\text{logit}\{P(M^*=1|M=m,\boldsymbol{Z};\boldsymbol{\gamma})\} = \gamma_{1m0} + \boldsymbol{\gamma_{1mZ}} \boldsymbol{Z}$

**Outcome mechanism:** $E(Y|X,\boldsymbol{C},M) = \theta_0 + \theta_X X + \boldsymbol{\theta_C C} + \theta_M M + \theta_{XM} XM$

#1: OLS Correction | #2: Predictive value weighting | **#3: An EM algorithm**

**Complete data log-likelihood:**

$$\ell_{complete}(\boldsymbol{\beta},\boldsymbol{\gamma},\boldsymbol{\theta};X,\boldsymbol{C},\boldsymbol{Z},Y)$$

$$= \sum_{i=1}^{N}\left[\ell_{Y|X,M,\boldsymbol{C}}(\boldsymbol{\theta};X_i,M_i,\boldsymbol{C}_i,Y_i) + \sum_{j=1}^{2} m_{ij}\log\{\pi_{ij}\} + \sum_{j=1}^{2}\sum_{\ell=1}^{2} m_{ij}m^*_{i\ell}\log\{\pi^*_{i\ell j}\}\right]$$

$\mathbb{I}(M_i = j)$ $P(M_i = j)$ $\mathbb{I}(M_i^* = \ell)$ $P(M_i^* = \ell \mid M_i = j)$

Outcome | True mediator | Observed mediator

2

# We developed 3 estimation methods

**True mediator mechanism:** $\text{logit}\{P(M=1|X,\boldsymbol{C};\boldsymbol{\beta})\} = \beta_0 + \beta_X X + \boldsymbol{\beta_C C}$

**Observed mediator mechanism:** $\text{logit}\{P(M^*=1|M=m,\boldsymbol{Z};\boldsymbol{\gamma})\} = \gamma_{1m0} + \boldsymbol{\gamma_{1mZ} Z}$

**Outcome mechanism:** $E(Y|X,\boldsymbol{C},M) = \theta_0 + \theta_X X + \boldsymbol{\theta_C C} + \theta_M M + \theta_{XM} XM$

#1: OLS Correction

#2: Predictive value weighting

#3: An EM algorithm

**Expectation Step** ⟷ **Maximization Step**

2

# We developed 3 estimation methods

**True mediator mechanism:** $\text{logit}\{P(M=1|X,\boldsymbol{C};\boldsymbol{\beta})\} = \beta_0 + \beta_X X + \boldsymbol{\beta_C C}$

**Observed mediator mechanism:** $\text{logit}\{P(M^*=1|M=m,\boldsymbol{Z};\boldsymbol{\gamma})\} = \gamma_{1m0} + \boldsymbol{\gamma_{1mZ}Z}$

**Outcome mechanism:** $E(Y|X,\boldsymbol{C},M) = \theta_0 + \theta_X X + \boldsymbol{\theta_C C} + \theta_M M + \theta_{XM} XM$

#1: OLS Correction | #2: Predictive value weighting | #3: An EM algorithm

**Expectation Step** ⟷ **Maximization Step**

$$w_{ij} = P(M_i = j|M_i^*, X_i, \boldsymbol{C}_i, \boldsymbol{Z}_i, Y_i)$$

$$= \sum_{\ell=1}^{2} \frac{m_{i\ell}^* \pi_{i\ell j}^* \pi_{ij} E[Y_i|X_i, M_i = j, \boldsymbol{C}_i, \boldsymbol{\theta}^{(t)}]}{\sum_{k=1}^{2} \pi_{i\ell k}^* \pi_{ik} E[Y_i|X_i, M_i = k, \boldsymbol{C}_i, \boldsymbol{\theta}^{(t)}]}$$

# We developed 3 estimation methods

**True mediator mechanism:** $\text{logit}\{P(M=1|X,\boldsymbol{C};\boldsymbol{\beta})\} = \beta_0 + \beta_X X + \boldsymbol{\beta_C C}$

**Observed mediator mechanism:** $\text{logit}\{P(M^*=1|M=m,\boldsymbol{Z};\boldsymbol{\gamma})\} = \gamma_{1m0} + \boldsymbol{\gamma_{1mZ}} \boldsymbol{Z}$

**Outcome mechanism:** $E(Y|X,\boldsymbol{C},M) = \theta_0 + \theta_X X + \boldsymbol{\theta_C C} + \theta_M M + \theta_{XM} XM$

#1: OLS Correction | #2: Predictive value weighting | #3: An EM algorithm

**Expectation Step** ⟷ **Maximization Step**

$$w_{ij} = P(M_i = j|M_i^*, X_i, \boldsymbol{C}_i, \boldsymbol{Z}_i, Y_i)$$
$$= \sum_{\ell=1}^{2} \frac{m_{i\ell}^* \pi_{i\ell j}^* \pi_{ij} E[Y_i|X_i, M_i = j, \boldsymbol{C}_i, \boldsymbol{\theta}^{(t)}]}{\sum_{k=1}^{2} \pi_{i\ell k}^* \pi_{ik} E[Y_i|X_i, M_i = k, \boldsymbol{C}_i, \boldsymbol{\theta}^{(t)}]}$$

$$Q = \sum_{i=1}^{N} \Big[ \sum_{j=1}^{2} \ell_{Y|X,M,\boldsymbol{C}}(\boldsymbol{\theta}; X_i, M_i = w_{ij}, \boldsymbol{C}_i, Y_i) + \sum_{j=1}^{2} w_{ij} \log\{\pi_{ij}\} + \sum_{j=1}^{2} \sum_{\ell=1}^{2} w_{ij} m_{i\ell}^* \log\{\pi_{i\ell j}^*\} \Big]$$

# We developed 3 estimation methods

**True mediator mechanism:** $\text{logit}\{P(M=1|X,\boldsymbol{C};\boldsymbol{\beta})\} = \beta_0 + \beta_X X + \boldsymbol{\beta_C C}$

**Observed mediator mechanism:** $\text{logit}\{P(M^*=1|M=m,\boldsymbol{Z};\boldsymbol{\gamma})\} = \gamma_{1m0} + \boldsymbol{\gamma_{1mZ}} \boldsymbol{Z}$

**Outcome mechanism:** $E(Y|X,\boldsymbol{C},M) = \theta_0 + \theta_X X + \boldsymbol{\theta_C C} + \theta_M M + \theta_{XM} XM$

#1: OLS Correction

#2: Predictive value weighting

#3: An EM algorithm

**Expectation Step**

Apply label switching correction from Webb and Wells (2023)

**Maximization Step**

$$w_{ij} = P(M_i = j|M_i^*, X_i, \boldsymbol{C}_i, \boldsymbol{Z}_i, Y_i)$$

$$= \sum_{\ell=1}^{2} \frac{m_{i\ell}^* \pi_{i\ell j}^* \pi_{ij} E[Y_i|X_i, M_i = j, \boldsymbol{C}_i, \boldsymbol{\theta}^{(t)}]}{\sum_{k=1}^{2} \pi_{i\ell k}^* \pi_{ik} E[Y_i|X_i, M_i = k, \boldsymbol{C}_i, \boldsymbol{\theta}^{(t)}]}$$

$$Q = \sum_{i=1}^{N} \Big[ \sum_{j=1}^{2} \ell_{Y|X,M,\boldsymbol{C}}(\boldsymbol{\theta}; X_i, M_i = w_{ij}, \boldsymbol{C}_i, Y_i) + \sum_{j=1}^{2} w_{ij} \log\{\pi_{ij}\} + \sum_{j=1}^{2} \sum_{\ell=1}^{2} w_{ij} m_{i\ell}^* \log\{\pi_{i\ell j}^*\} \Big]$$

2

# We developed 3 estimation methods

**True mediator mechanism:** $\text{logit}\{P(M=1|X,\boldsymbol{C};\boldsymbol{\beta})\} = \beta_0 + \beta_X X + \boldsymbol{\beta_C C}$

**Observed mediator mechanism:** $\text{logit}\{P(M^*=1|M=m,\boldsymbol{Z};\boldsymbol{\gamma})\} = \gamma_{1m0} + \boldsymbol{\gamma_{1mZ} Z}$

**Outcome mechanism:** $E(Y|X,\boldsymbol{C},M) = \theta_0 + \theta_X X + \boldsymbol{\theta_C C} + \theta_M M + \theta_{XM} XM$

| #1: OLS Correction | #2: Predictive value weighting | #3: An EM algorithm |
|---|---|---|

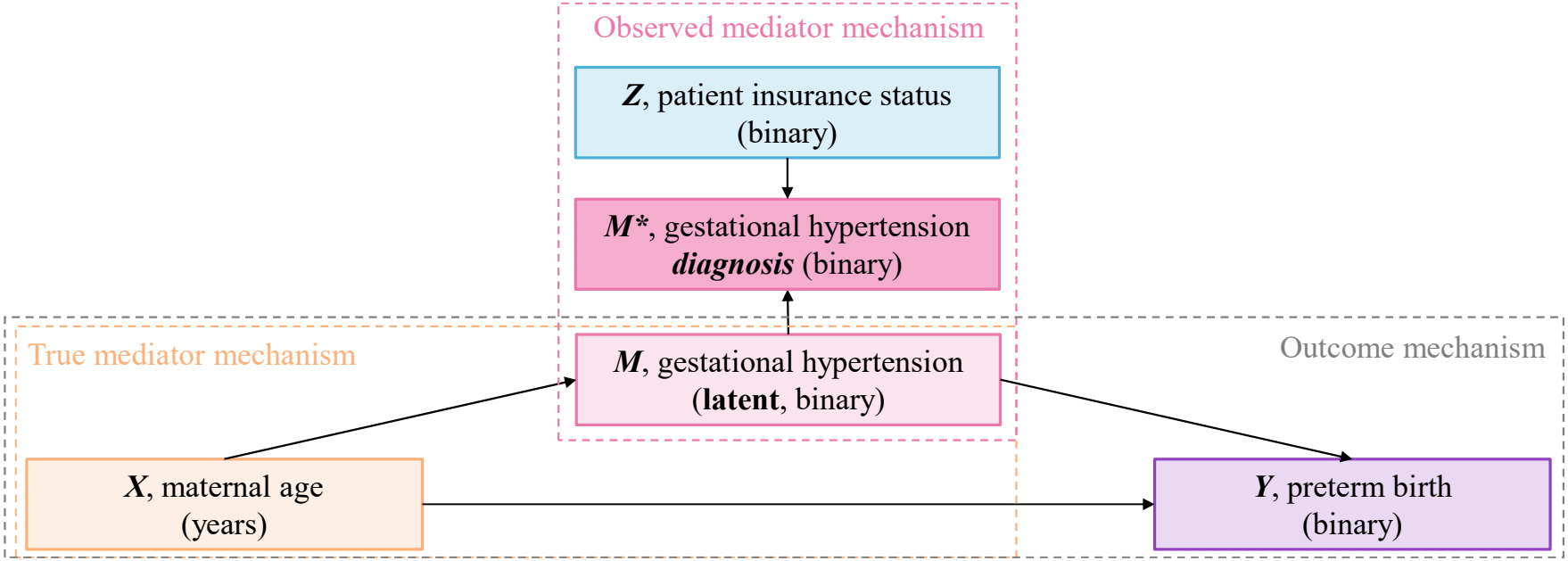
- Use the resulting bias-corrected parameter estimates to compute **(in)direct effects** for a change from **x̃ to x:**

$$OR^{NDE} \cong \frac{\exp(\theta_X x)\left\{1+\exp(\theta_M + \theta_{XM}x + \beta_0 + \beta_X\tilde{x} + \beta_C c)\right\}}{\exp(\theta_X \tilde{x})\left\{1+\exp(\theta_M + \theta_{XM}\tilde{x} + \beta_0 + \beta_X\tilde{x} + \beta_C c)\right\}}$$

$$OR^{NIE} \cong \frac{\left\{1+\exp(\beta_0 + \beta_X\tilde{x} + \beta_C c)\right\} 1+\exp(\theta_X + \theta_{XM}x + \beta_0 + \beta_X x + \beta_C c)\}}{\left\{1+\exp(\beta_0 + \beta_X x + \beta_C c)\right\}\left\{1+\exp(\theta_M + \theta_{XM}x + \beta_0 + \beta_X\tilde{x} + \beta_C c)\right\}}$$

# We applied our methods to a preterm birth study

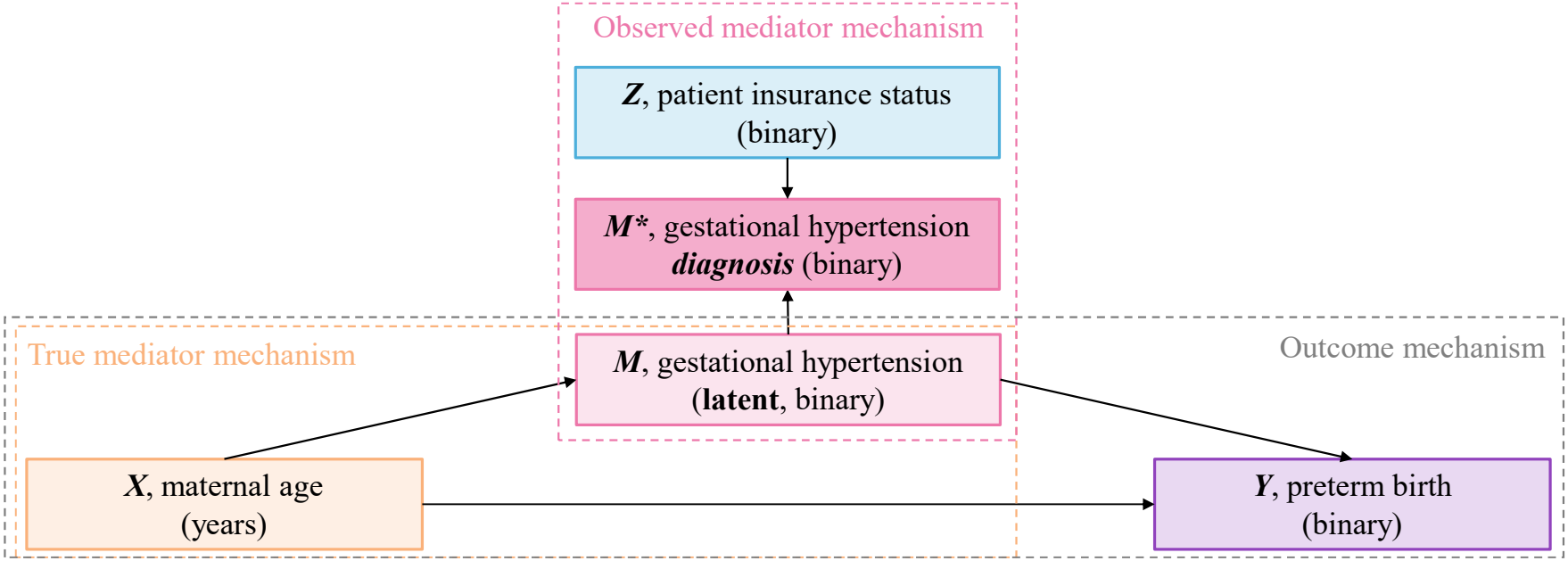
? Does **gestational hypertension** mediate the association between **maternal age** and **preterm birth**, after accounting for potential **misdiagnosis of gestational hypertension** based on **patient insurance status**?



# We applied our methods to a preterm birth study

**Data:** National Vital Statistics System

- Provides demographic and health data for all births in a year in the US.
- Random subsample from calendar year 2021, **N = 20,000**.

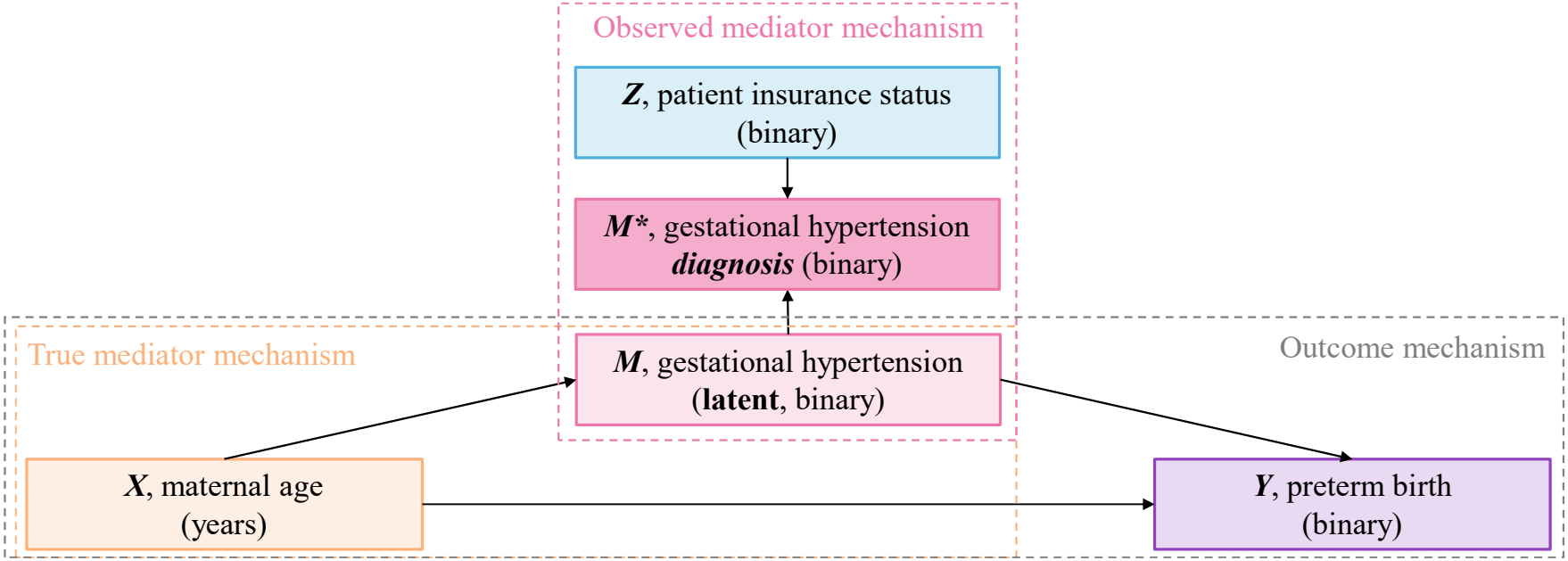


# We applied our methods to a preterm birth study

**True mediator mechanism:** **M** ~ **X** + Race + Education + Parity + Smoking + BMI

**Observed mediator mechanism:** **M*** | **M** ~ Race + **Z**

**Outcome mechanism:** **Y** ~ **X** + Race + Education + Parity + Smoking + BMI + **M** + **M** * **X**



# Results change when we account for misdiagnosis

**True mediator mechanism:** **M** ~ **X** + Race + Education + Parity + Smoking + BMI

**Observed mediator mechanism:** **M* | M** ~ Race + **Z**

**Outcome mechanism:** **Y** ~ **X** + Race + Education + Parity + Smoking + BMI + **M** + **M * X**

| | **EM Algorithm** | | **Naïve Analysis** | |
|---|---|---|---|---|
| | Est. | SE | Est. | SE |
| $\beta_X$ | | | | |
| $\gamma_{Z,\ G = 1}$ | | | | |
| $\gamma_{Z,\ G = 2}$ | | | | |
| $\theta_X$ | | | | |
| $\theta_M$ | | | | |
| $\theta_{XM}$ | | | | |

3

# Results change when we account for misdiagnosis

**True mediator mechanism:** **M** ~ **X** + Race + Education + Parity + Smoking + BMI

**Observed mediator mechanism:** **M*** | **M** ~ Race + **Z**

**Outcome mechanism:** **Y** ~ **X** + Race + Education + Parity + Smoking + BMI + **M** + **M * X**

Association between **age** & **gestational hypertension** is unchanged after accounting for misdiagnosis.

| | **EM Algorithm** | | **Naïve Analysis** | |
|---|---|---|---|---|
| | Est. | SE | Est. | SE |
| $\beta_X$ | 0.10 | 0.04 | 0.08 | 0.03 |
| $\gamma_{Z,\ G = 1}$ | | | | |
| $\gamma_{Z,\ G = 2}$ | | | | |
| $\theta_X$ | | | | |
| $\theta_M$ | | | | |
| $\theta_{XM}$ | | | | |

3

# Results change when we account for misdiagnosis

**True mediator mechanism:** **M** ~ **X** + Race + Education + Parity + Smoking + BMI

**Observed mediator mechanism:** **M* | M** ~ Race + **Z**

**Outcome mechanism:** **Y** ~ **X** + Race + Education + Parity + Smoking + BMI + **M** + **M * X**

Association between **age** & **gestational hypertension** is unchanged after accounting for misdiagnosis.

Association between **gestational hypertension** & **preterm birth** strengthens.

| | **EM Algorithm** | | **Naïve Analysis** | |
|---|---|---|---|---|
| | Est. | SE | Est. | SE |
| $\beta_X$ | 0.10 | 0.04 | 0.08 | 0.03 |
| $\gamma_{Z,\ G=1}$ | | | | |
| $\gamma_{Z,\ G=2}$ | | | | |
| $\theta_X$ | 0.02 | 0.05 | 0.10 | 0.03 |
| $\theta_M$ | 1.19 | 0.17 | 0.88 | 0.06 |
| $\theta_{XM}$ | 0.19 | 0.09 | 0.06 | 0.06 |

3

# Results change when we account for misdiagnosis

**True mediator mechanism:** **M** ~ **X** + Race + Education + Parity + Smoking + BMI

**Observed mediator mechanism:** **M*** | **M** ~ Race + **Z**

**Outcome mechanism:** **Y** ~ **X** + Race + Education + Parity + Smoking + BMI + **M** + **M * X**

Association between **age** & **gestational hypertension** is unchanged after accounting for misdiagnosis.

Association between **gestational hypertension** & **preterm birth** strengthens.

| | **EM Algorithm** | | **Naïve Analysis** | |
|---|---|---|---|---|
| | Est. | SE | Est. | SE |
| $\beta_X$ | 0.10 | 0.04 | 0.08 | 0.03 |
| $\gamma_{Z,\, G = 1}$ | -1.01 | 0.40 | - | - |
| $\gamma_{Z,\, G = 2}$ | 2.09 | 8.81 | - | - |
| $\theta_X$ | 0.02 | 0.05 | 0.10 | 0.03 |
| $\theta_M$ | 1.19 | 0.17 | 0.88 | 0.06 |
| $\theta_{XM}$ | 0.19 | 0.09 | 0.06 | 0.06 |

Use γ estimates to compute **sensitivity and specificity**.

3

# *M* is measured with perfect specificity and low sensitivity

**True mediator mechanism:** **M** ~ **X** + Race + Education + Parity + Smoking + BMI

**Observed mediator mechanism:** **M*** | **M** ~ Race + **Z**

**Outcome mechanism:** **Y** ~ **X** + Race + Education + Parity + Smoking + BMI + **M** + **M** * **X**

| | **Estimated Specificity**<br>P( no M* \| no M ) | **Estimated Sensitivity**<br>P( M* \| M ) |
|---|---|---|
| Insured | 99.9% | 43.1% |
| Self-Pay | 99.4% | 21.7% |

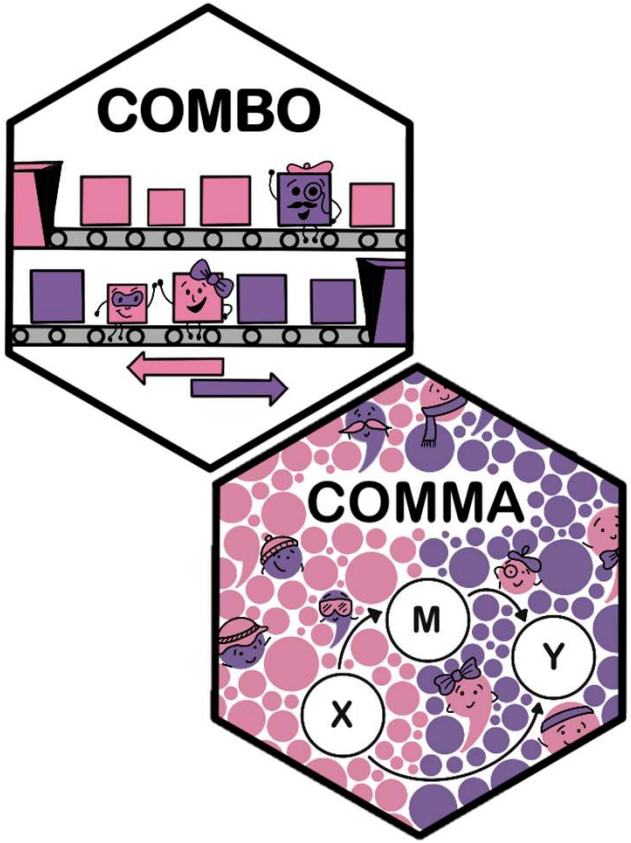
# Key takeaways

- Developed new methods for handling misclassified binary mediator variables.
- Computed (in)direct effects using bias-corrected parameter estimates.
- Quantified gestational hypertension misdiagnosis rates based on insurance status.

# Software

- Estimation methods for **misclassified outcomes** are available in the *COMBO* R Package on CRAN.
  - **Co**rrecting **M**isclassified **B**inary **O**utcomes

- Estimation methods for **misclassified mediators** are available in the *COMMA* R Package on CRAN.
  - **Co**rrecting **M**isclassified **M**ediation **A**nalysis



# Thank you!

**Kimberly A. H. Webb**

kimberlywebb@pitt.edu

kimhwebb.com → My "webb-site" ☺

Scan for the paper

bit.ly/Webb-SMMR

University of Pittsburgh

Cornell Bowers CIS
Statistics and Data Science